Title: Hydrometallurgical Metal Extraction Process



Figure 1: The Complete Hydrometallurgical Process – Original Six-Step Sequence

Figure 1 illustrates the complete hydrometallurgical process used to extract and recover metals from ores or waste materials through a series of aqueous chemical steps. The process begins with Ore Preparation, in which the ore or waste feed is crushed and ground to increase surface area and liberate the valuable minerals. In the Leaching step, the prepared material is treated with a suitable leaching agent – an acid, alkali, or salt solution – that dissolves the desired metal to form a metal-rich solution. Solid–Liquid Separation then removes the remaining solid residue from the leach solution by filtration, settling, or centrifugation, producing a clear solution containing the dissolved metals. During Solution Purification, impurities are removed from the leach solution using methods such as precipitation, solvent extraction, or ion exchange, yielding a purified metal solution. In Metal Recovery, the pure metal is recovered from the solution using electrowinning, chemical precipitation, or cementation. Finally, Refining further purifies the recovered metal to achieve the purity and quality standards required for industrial use.

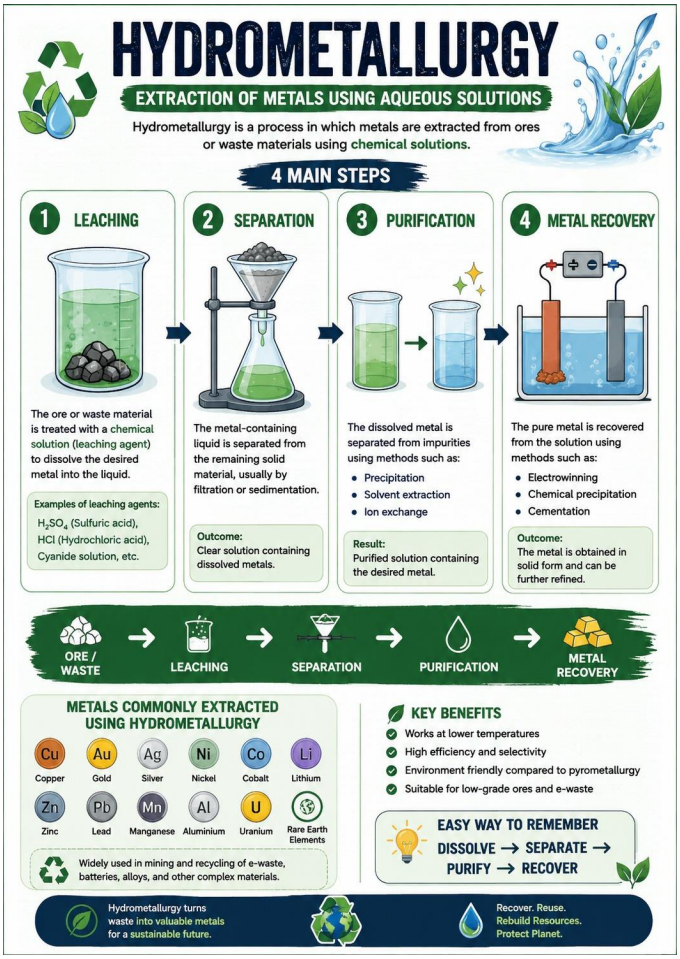


Figure 2: The Reduced Process – Simplified Four-Step Sequence

Figure 2 illustrates the same hydrometallurgical process consolidated into four main steps for ease of reference. The initial ore or waste feed is treated directly with a chemical leaching solution – such as sulfuric acid, hydrochloric acid, or a cyanide solution – in the Leaching step, dissolving the desired metal into the liquid. The Separation step removes the metal-containing liquid from the remaining solid material, typically by filtration or sedimentation, producing a clear solution containing the dissolved metals. Purification then separates the dissolved metal from impurities using precipitation, solvent extraction, or ion exchange, resulting in a purified solution containing the desired metal. In the final Metal Recovery step, the pure metal is recovered from the solution by electrowinning, chemical precipitation, or cementation, and is obtained in solid form ready for further refining. This reduced sequence groups ore preparation into the leaching stage and treats refining as an optional finishing stage, retaining the same chemical logic as the original six-step process while presenting it more concisely.

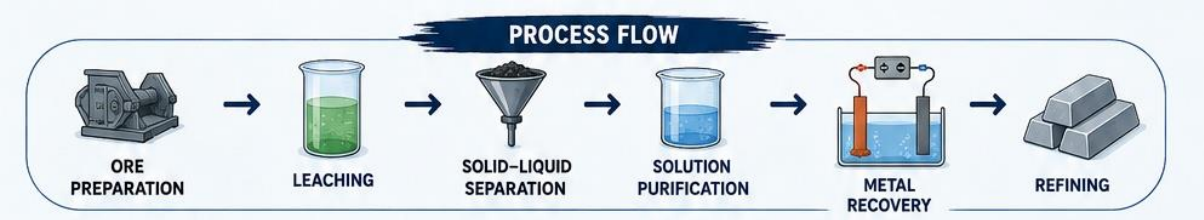


Figure 3: Process Flow Diagram – Six-Step Sequence

Figure 3 illustrates the process flow of the complete six-step hydrometallurgical sequence: Ore Preparation → Leaching → Solid–Liquid Separation → Solution Purification → Metal Recovery → Refining. This flow is easily remembered by the sequence Prepare → Leach →

Separate → Purify → Recover → Refine, with each stage feeding directly into the next until the metal reaches the required purity for industrial use.



Figure 4: Process Flow Diagram – Reduced Four-Step Sequence

Figure 4 illustrates the process flow of the reduced four-step sequence: Ore/Waste → Leaching → Separation → Purification → Metal Recovery, remembered by the shorter sequence Dissolve → Separate → Purify → Recover. Hydrometallurgy of this kind operates at comparatively low temperatures, offers high efficiency and selectivity, and is environmentally favourable compared with pyrometallurgy, making it well suited to low-grade ores and e-waste. It is widely applied in the mining and recycling of ores, e-waste, batteries, and industrial by-products to recover metals such as copper, gold, silver, nickel, cobalt, lithium, zinc, lead, manganese, aluminium, uranium, and rare earth elements.

This document consolidates standard hydrometallurgical process knowledge, presenting both the complete six-step sequence and its condensed four-step equivalent side by side for clarity and ease of reference.